Network Security Repeat CA Assignment

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Module: Network Security
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Declaration:

I declare that this report and the associated Packet Tracer implementation are my own work. Any sources or module materials used during the project have been acknowledged in the references section.

Signed: Glory Isoken Oladipo Omoyemwense

Date: 19 August 2026

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1. Introduction

This project focused on improving the security of an existing network infrastructure consisting of three sites: Dublin Headquarters, the London Data Centre and the Beijing Home Office. The original network provided connectivity between the sites but required additional security controls to protect network devices, users and data.

I implemented a range of security measures across the network, including device hardening, secure remote access and AAA authentication, centralised logging and time synchronisation, OSPF authentication, packet-filtering access control lists, LAN and wireless security, and Port Address Translation (PAT). Each implemented control was tested where it was possible to verify that it works as it should. Some additional requirements could not be fully implemented and are discussed later in the report.

2. Network Device Hardening

2.1 Implementation and Purpose

Network device hardening was conducted on the Dublin and London routers to improve the security of administrative access to network devices. Password protection, encrypted credentials, minimum password requirements, warning banners and session timeouts were also configured. SSH version 2 was also used for secure remote management.

These types of controls reduce the chances of unauthorised access to network infrastructures. Password security protects administrative accounts, while session timeouts lower the chances of an authenticated session being left unattended and open. SSH offers encrypted remote administration rather than exposing management information over an unencrypted transport.

2.2 Configuration

```
The routers were configured using commands such as:

service password-encryption

security passwords min-length 8

enable secret ********

ip ssh version 2
```

A local administrator account was also created and an authorised-access MOTD message banner was configured for authorised access. Console and VTY lines were configured with inactivity timeouts to automatically terminate idle administrative sessions.

2.3 Testing and Evidence

The running configurations of the routers were reviewed to be sure that the hardening controls were successfully implemented.

Figure 1: Dublin router device-hardening configuration.

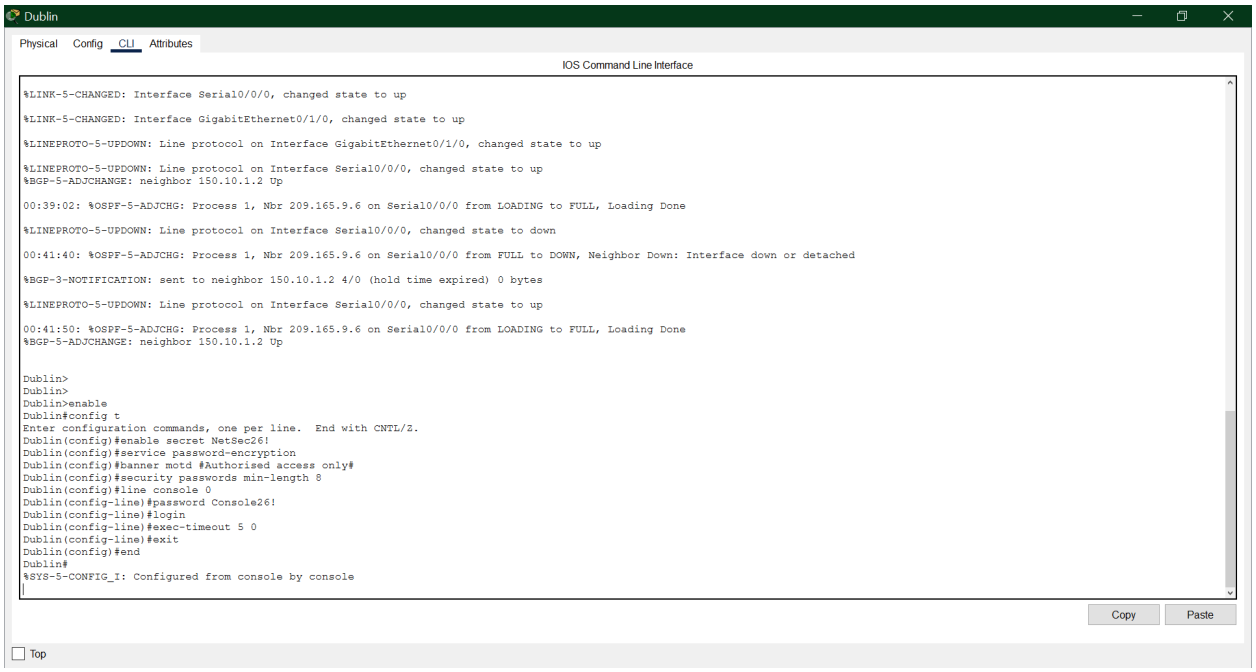
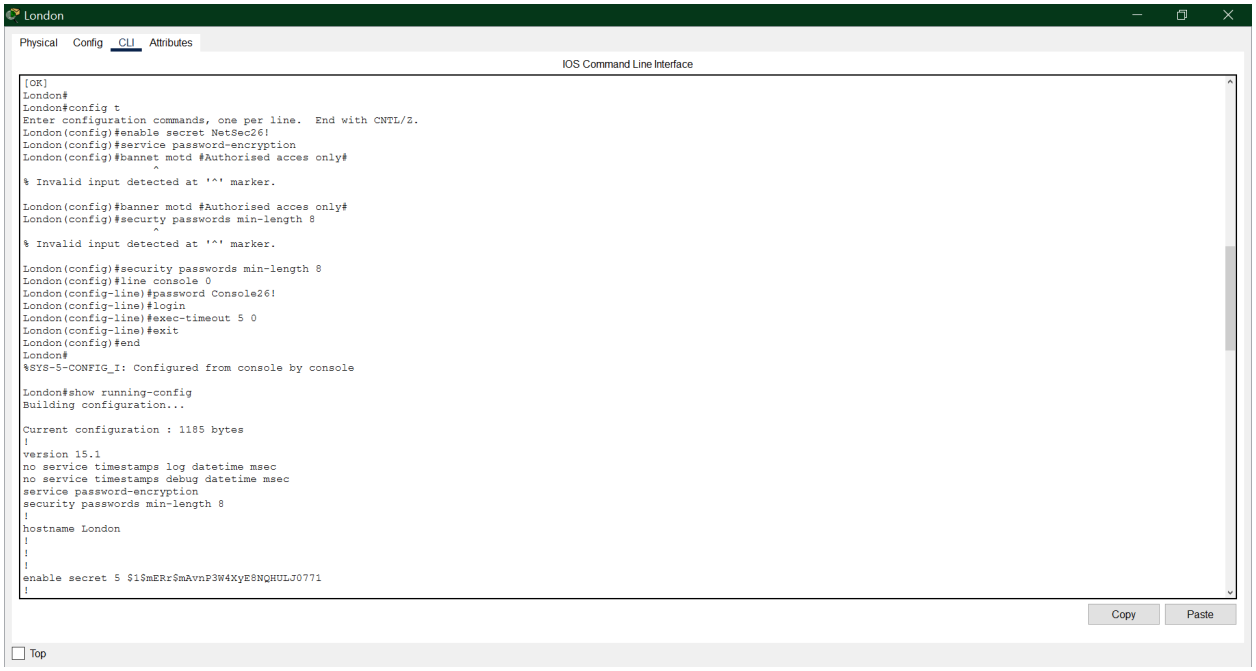


Figure 2: London router device-hardening configuration.



3. Secure Remote Access and AAA

3.1 Implementation and Purpose

Secure remote administration was configured using SSH. The London router was additionally set up to use AAA authentication through the RADIUS server at `192.168.3.6`. Local authentication was kept as a fallback if the RADIUS server became unavailable.

SSH secures the administrative connections by encrypting communication between the administrator and network device. AAA provides centralised authentication and also allows administrative access possible via an authenticating facility.

3.2 Configuration

AAA was enabled on the London router and the RADIUS server was specified as the primary authentication source. An AAA login technique was then used on the VTY lines.

```
Important configuration included:

aaa new-model

radius-server host 192.168.3.6 key ********

aaa authentication login AAA_LOGIN group radius local

The VTY lines were configured using:

login authentication AAA_LOGIN

transport input ssh
```

This restricted remote VTY access to SSH while using the configured AAA authentication method.

3.3 Testing and Evidence

The configuration was verified using `show running-config`. An SSH connection was then established from a device in the London Admin LAN to confirm that remote administration was functional.

Figure 3: London AAA and VTY configuration.

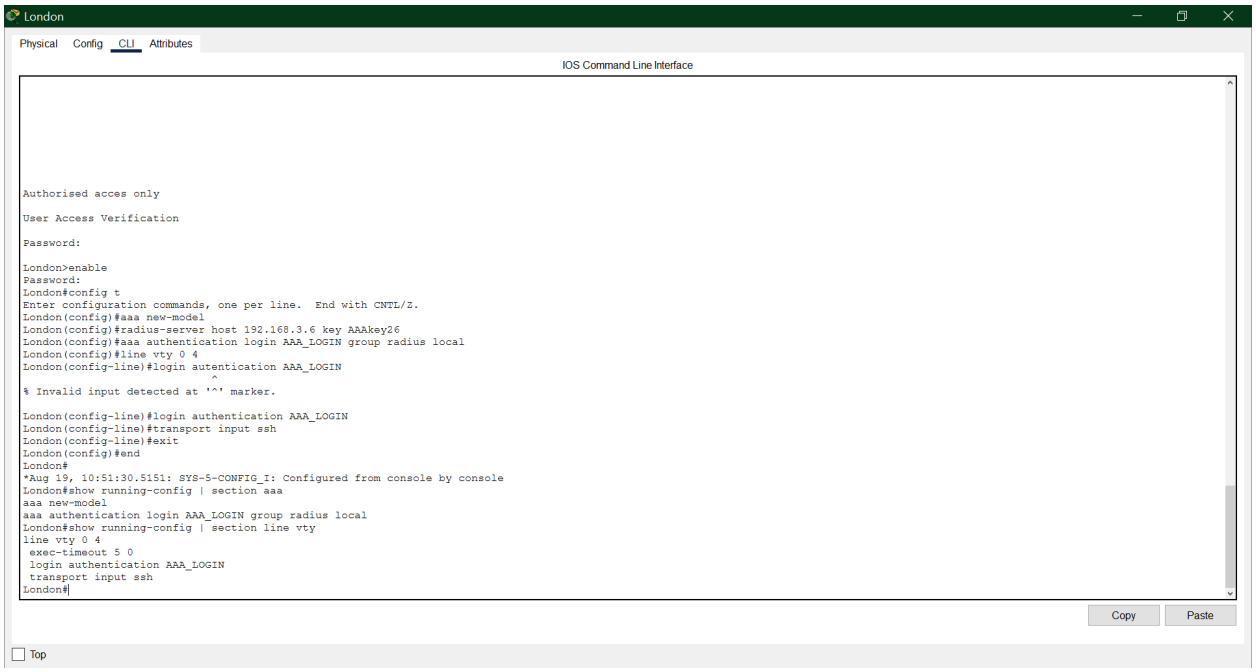
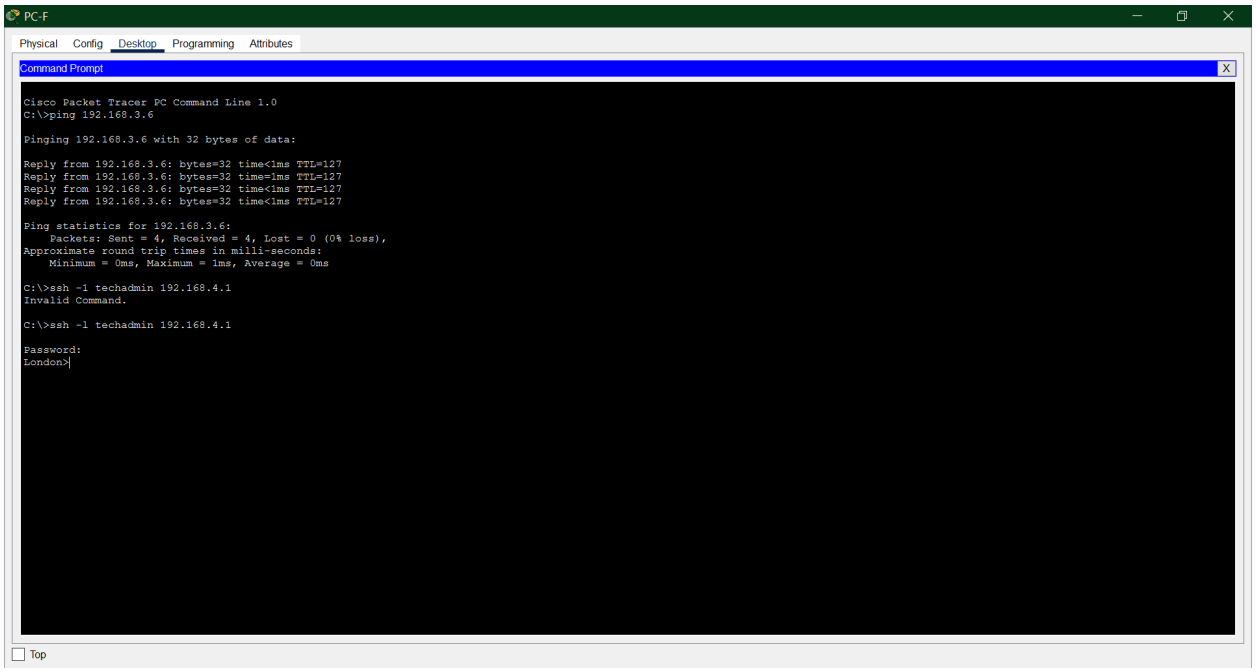


Figure 4: Successful SSH connection from the London Admin LAN.



4. Device Monitoring and Time Synchronisation

4.1 Implementation and Purpose

A central Syslog and NTP server was added to the London Server LAN using the address `192.168.3.5`. The Dublin and London routers were configured to use this server as their logging and time synchronisation.

Centralised logging enables the gathering of events in one location, making debugging and security monitoring easier. The NTP protocol helps provide accurate time so that events generated by different devices can be compared during an investigation.

4.2 Configuration

The routers were configured with the central server using commands including:

```
logging 192.168.3.5

ntp server 192.168.3.5

service timestamps log datetime msec

The Syslog and NTP services were enabled on the server.
```

4.3 Testing and Evidence

The logging configuration was checked through the `show logging` command which confirmed that messages were being sent to the destination `192.168.3.5`. NTP configuration was also verified by displaying the router's NTP associations.

Figure 5: Centralised Syslog configuration.

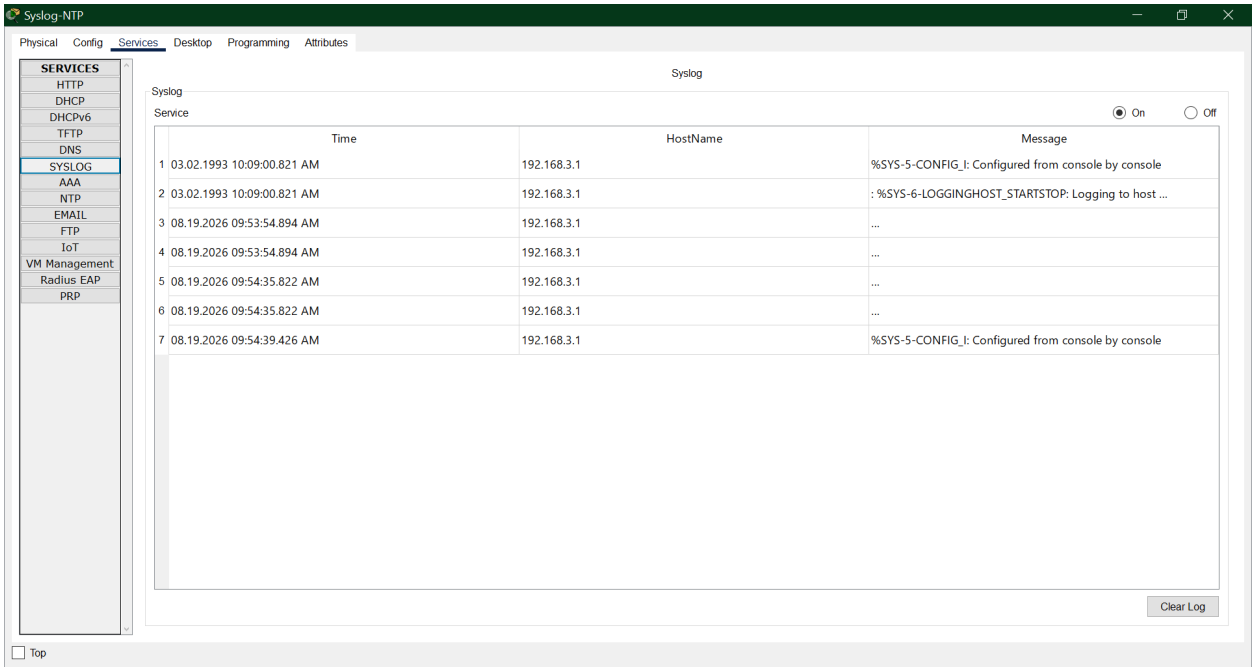
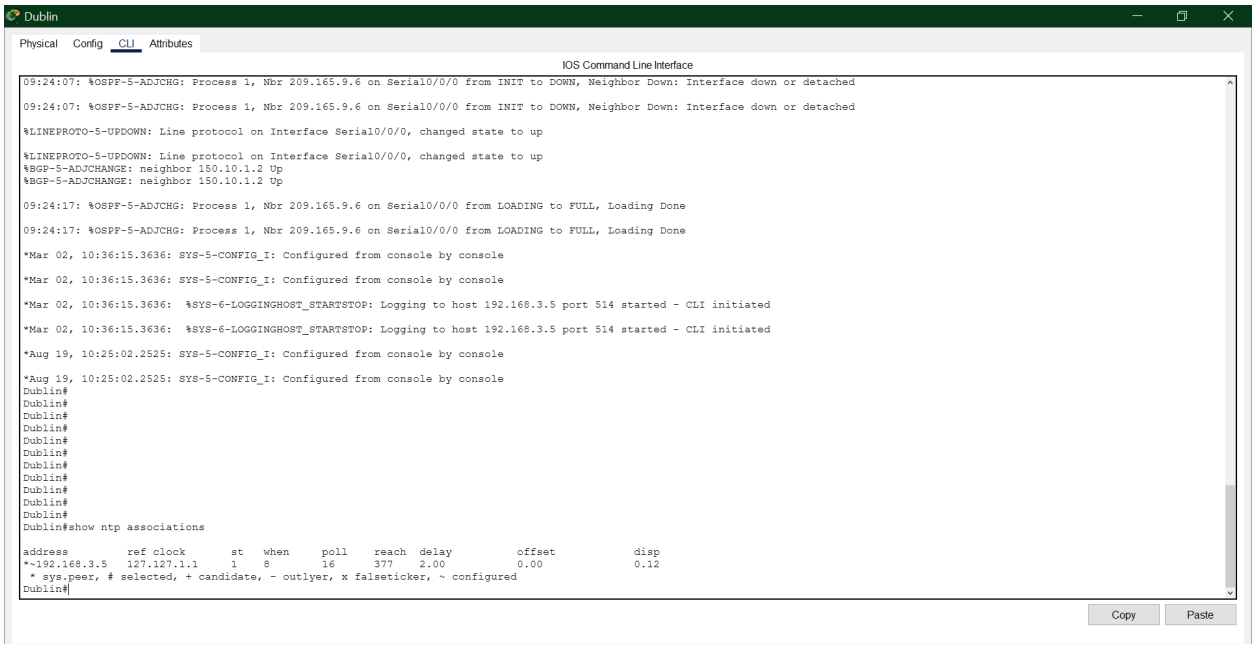


Figure 6: Dublin NTP association with the central server.



5. Control Plane Security — OSPF Authentication

5.1 Implementation and Purpose

OSPF authentication was configured between the Dublin and London routers to protect the communication of the routing information. The message-digest authentication was configured for the serial interfaces of the two routers.

An unauthorised device could potentially attempt to participate in the routing process or advertise false routing information if we didn't have the authentication configured in the routing process. OSPF authentication helps ensure that routing updates are accepted by the router only when the neighbor has the right credentials.

5.2 Configuration

The serial interfaces were configured using message-digest authentication:

```
ip ospf authentication message-digest

ip ospf message-digest-key 1 md5 *******
```

Matching authentication information was configured on both ends of the Dublin–London connection.

5.3 Testing and Evidence

During configuration, the OSPF adjacency was disrupted because matching authentication had not yet been configured on both ends. Once matching authentication was configured on both routers, the neighbour relationship reached the **FULL** state. This shows that routing communication was functioning with the authentication.

Figure 7: OSPF authentication configuration.

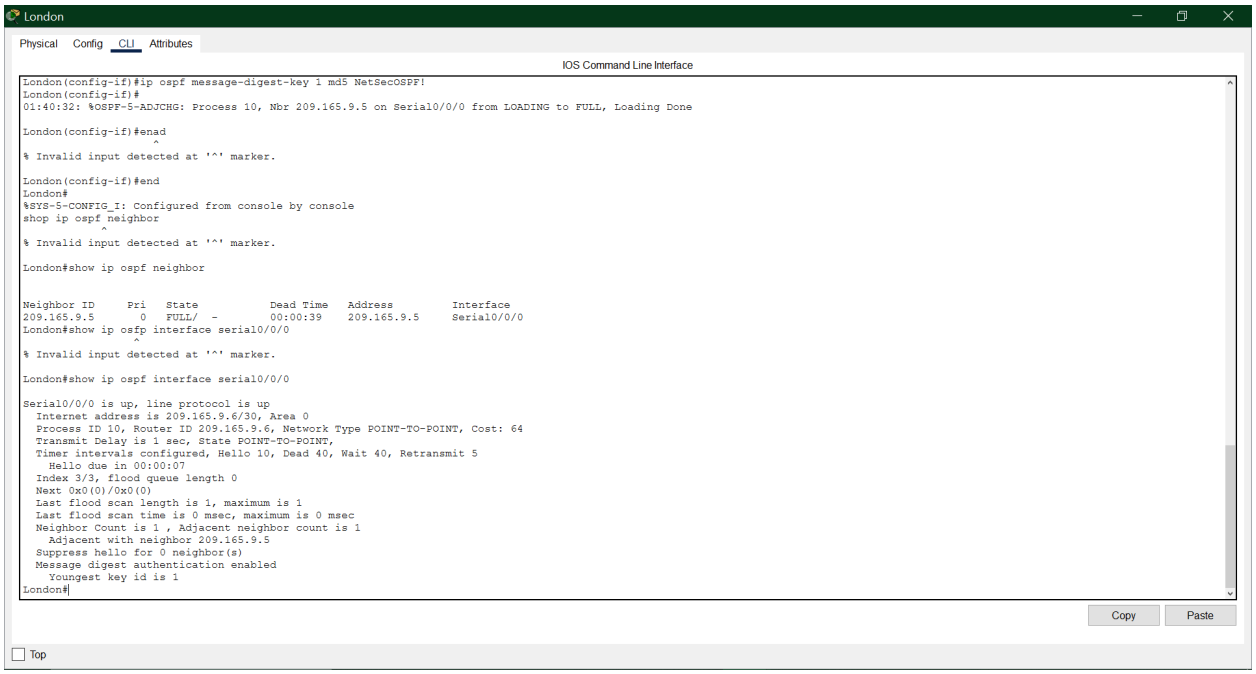
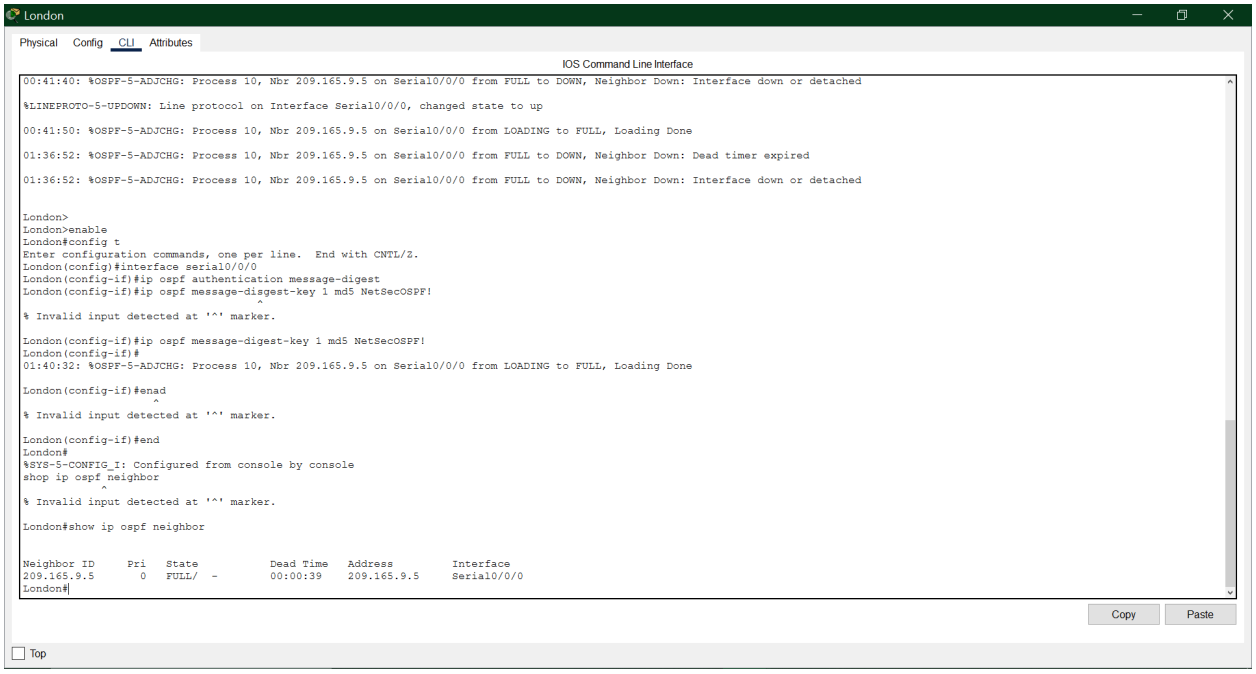


Figure 8: OSPF neighbour verified in FULL state.



6. Gateway Security — Packet Filtering

6.1 Implementation and Purpose

Packet-filtering ACLs were implemented on the Dublin router in order to isolate the Sales and Purchasing networks from each other. Sales operates on the `192.168.1.0/24` network on VLAN 10, while Purchasing uses `192.168.2.0/24` on VLAN 20.

Two extended ACLs were used so that communication was restricted in both directions. This is an example of network segmentation and can reduce lateral movement between departments if a host on one network becomes compromised.

6.2 Configuration

Traffic originating from Sales and destined for Purchasing was controlled using:

```
ip access-list extended SALE_FILTER

deny ip 192.168.1.0 0.0.0.255 192.168.2.0 0.0.0.255

permit ip any any
```

This ACL was applied inbound to subinterface G0/0.10.

The reverse direction was controlled using:

```
ip access-list extended PURCHASE_FILTER

deny ip 192.168.2.0 0.0.0.255 192.168.1.0 0.0.0.255

permit ip any any
```

This was applied inbound to `G0/0.20`.

6.3 Testing and Evidence

There was an attempt made by a Sales host at `192.168.1.3` to ping the Purchasing host at `192.168.2.2`. The test resulted in 100% packet loss. The reverse test from the Purchasing to Sales was also unsuccessful as expected.

The ACL was checked using `show access-lists`. The match counter on `SALE_FILTER` increased to four matches, confirming that the ICMP test packets had reached and matched the deny rule.

Figure 9: Dublin inter-VLAN ACL configuration and match counters.

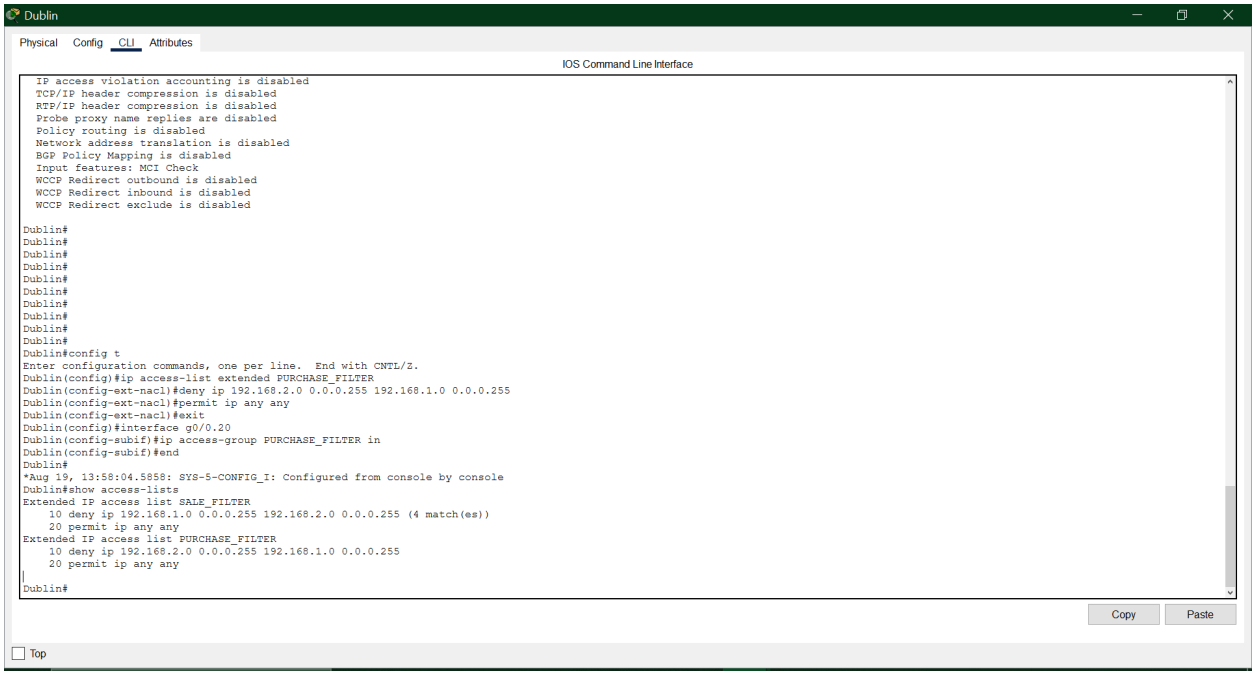
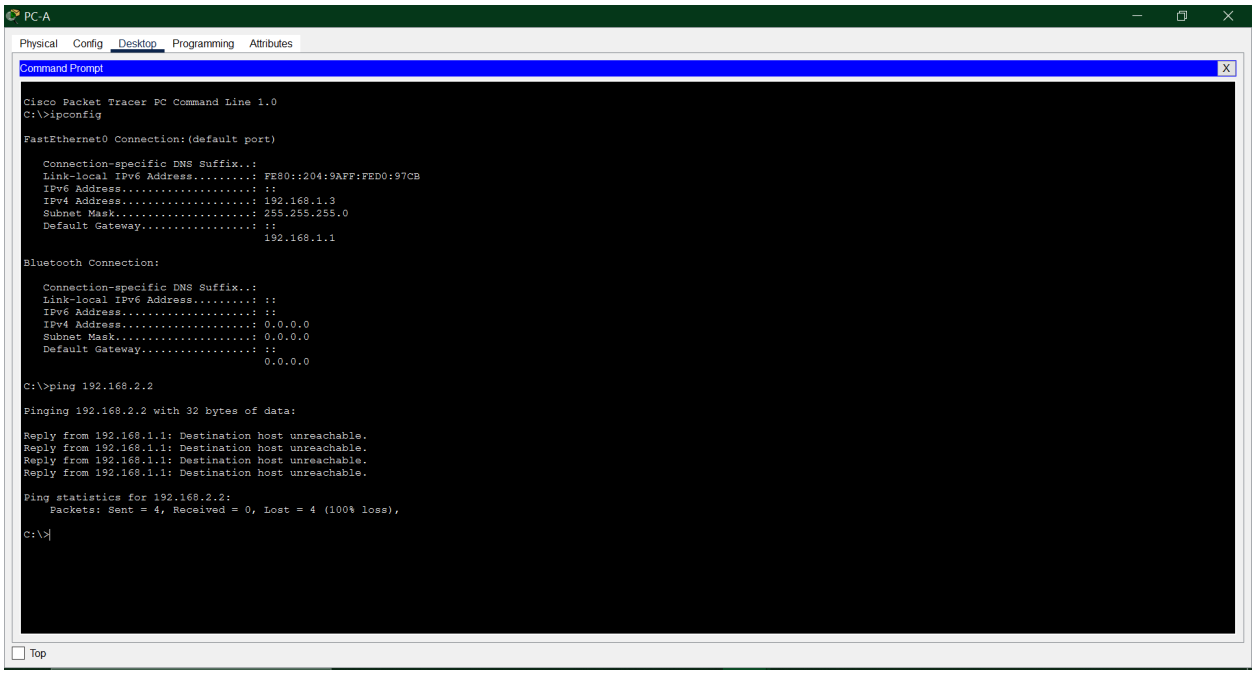


Figure 10: Sales-to-Purchasing traffic successfully blocked.



7. LAN Security

7.1 Implementation and Purpose

The layer 2 security controls were put in place on selected switch access ports. Port security was enabled, sticky MAC learning was used and the number of permitted MAC addresses was limited. Unused switch interfaces were also disabled.

Port security limits which devices can use an access port and can reduce the risk of unauthorised devices being connected to the LAN. Disabling the unused ports further minimizes the number of physical interfaces that could be used to gain network access.

7.2 Configuration

The configuration included commands such as:

```
switchport mode access

switchport port-security

switchport port-security maximum 1

switchport port-security mac-address sticky

switchport port-security violation restrict
```

Unused interfaces were placed into a shutdown state.

7.3 Testing and Evidence

Port-security configuration was checked on the switch to confirm that the security settings had been applied.

Figure 12: Switch port-security configuration.

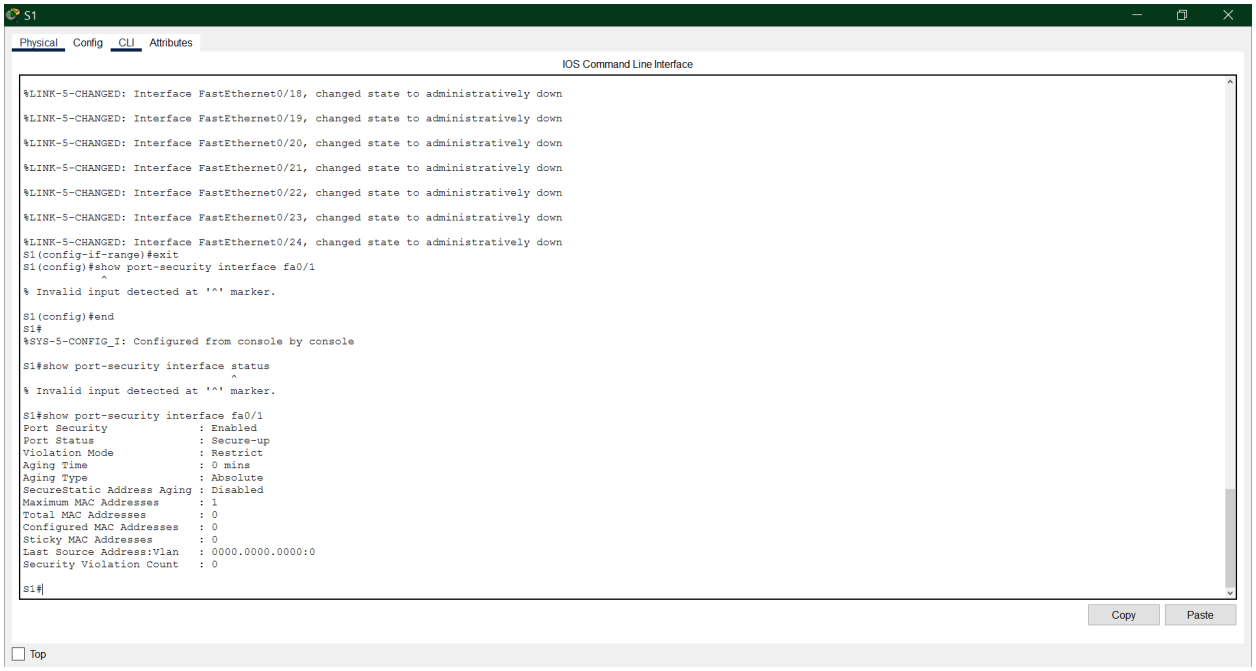
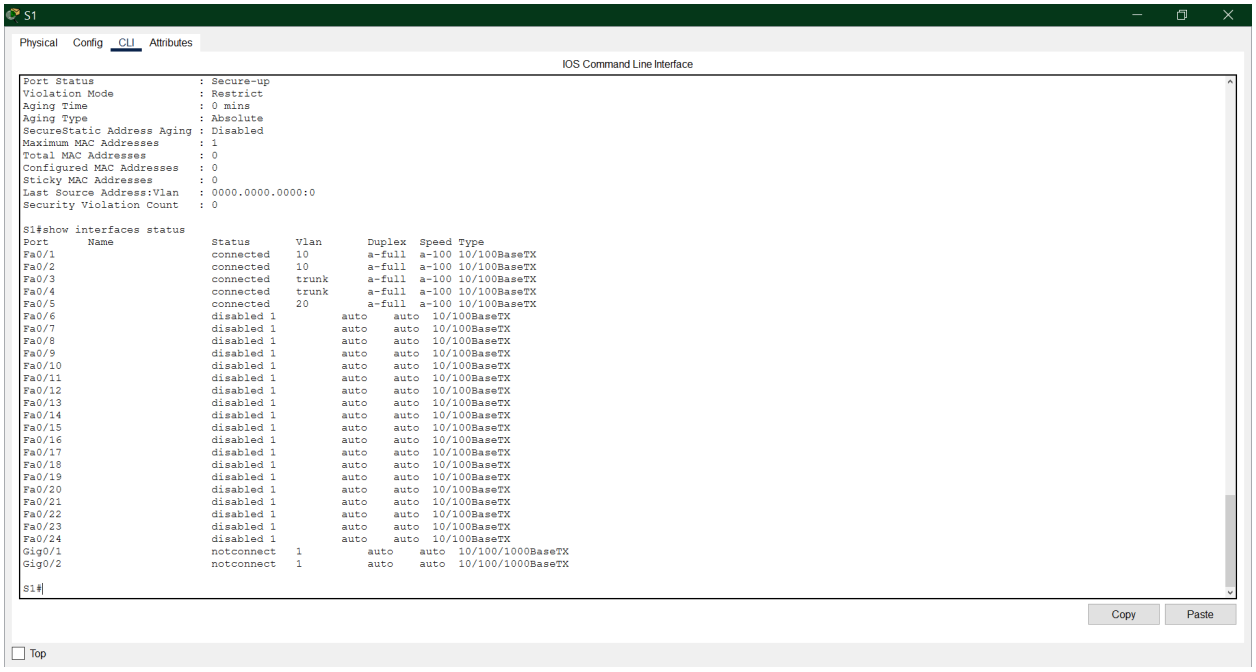


Figure 13: Unused switch interfaces disabled.



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8. WLAN Security — Beijing

8.1 Implementation and Purpose

The Beijing wireless network was protected by changing the default SSID with `TechNova-Beijing`, disabling SSID broadcast, enabling WPA2-Personal with AES encryption and configuring a pre-shared key. MAC-address filtering was also used to only permit specified wireless devices access to the network.

WPA2 with AES encrypts wireless communications and ensures that clients have proper credentials before joining the WLAN. MAC filtering provides an additional access-control since it limits access to wireless adapters.

8.2 MAC Address Filtering

The Wireless0 adapter information on PC-I was obtained using:

```
ipconfig /all
```

The adapter had the physical address:

```
0002.1686.30BE
```

This was entered into the wireless router's MAC filter in the required format:

```
00:02:16:86:30:BE
```

The filter was configured to permit listed devices to access the wireless network.

8.3 Testing and Troubleshooting

When the SSID and security settings were modified, PC-I temporarily lost wireless connectivity because its existing wireless configuration no longer matched the new WLAN settings. A first ping attempt failed.

When the wireless profile on PC-I was updated with the new SSID and WPA2 security information, the device then successfully reconnected to the access point, and a second ping to the Beijing gateway at `192.168.5.1` was successful.

This demonstrated both the impact of the modification of wireless security settings and successful connectivity had been achieved.

Figure 14: PC-I Wireless addressing and physical address.

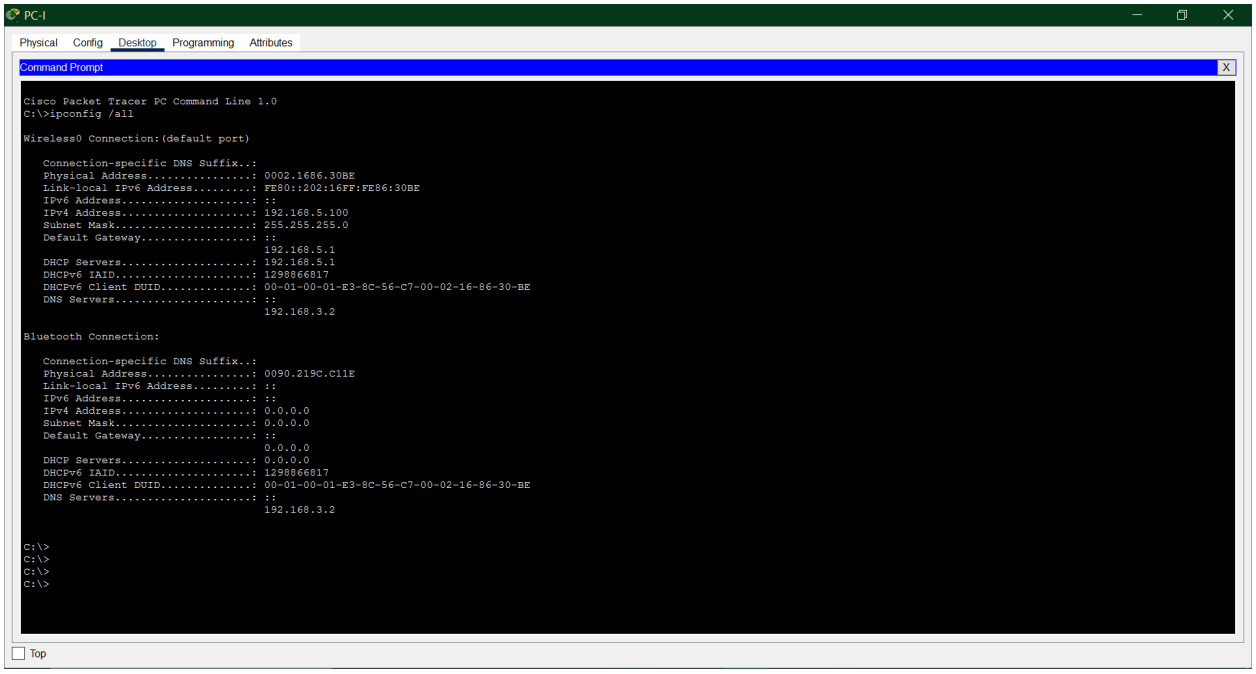


Figure 15: Beijing WPA2-Personal and AES configuration.

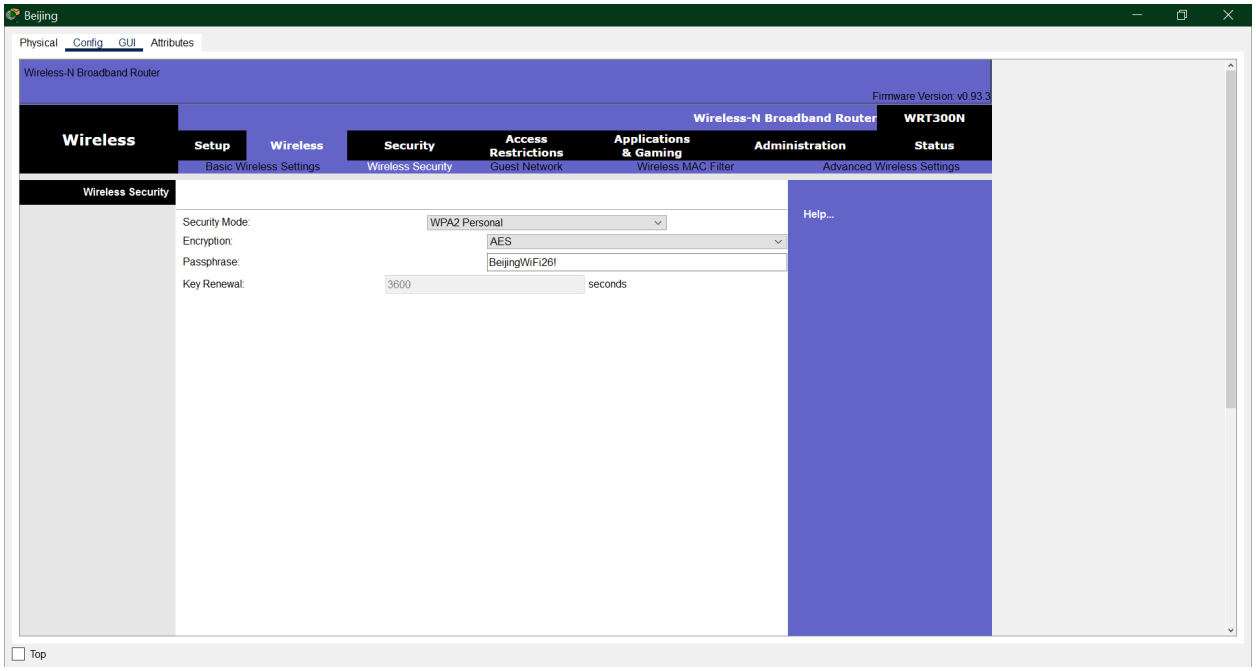


Figure 16: Beijing wireless MAC filtering configuration.

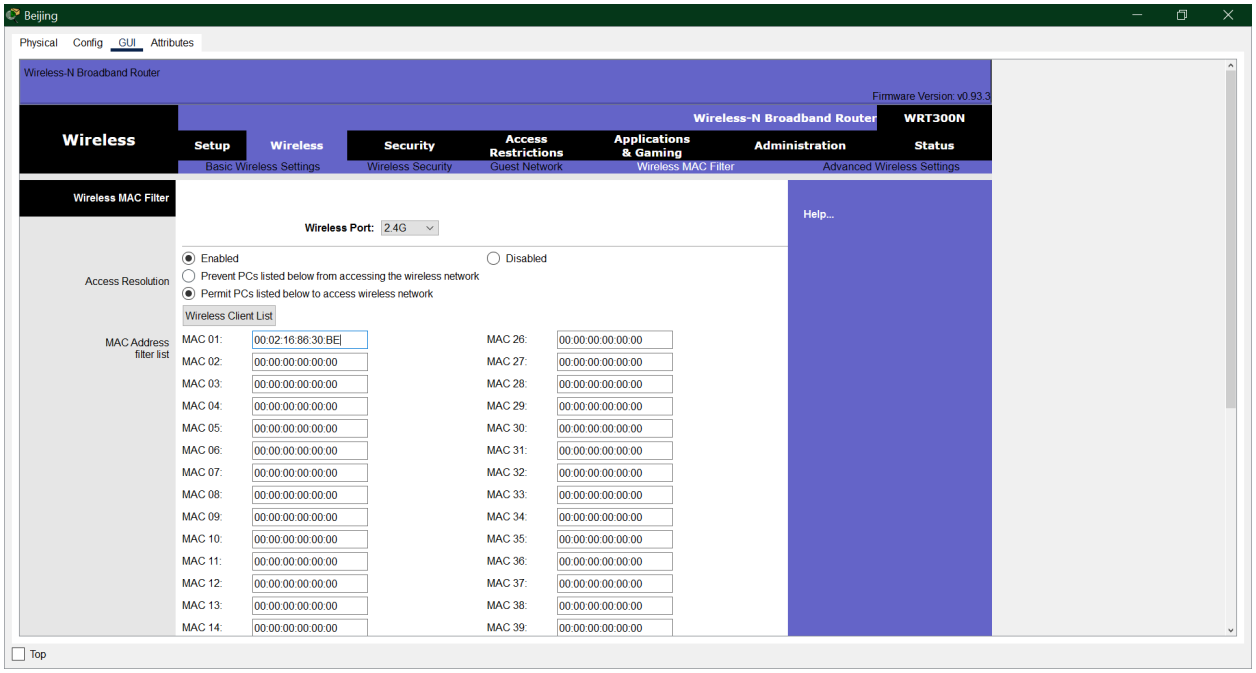
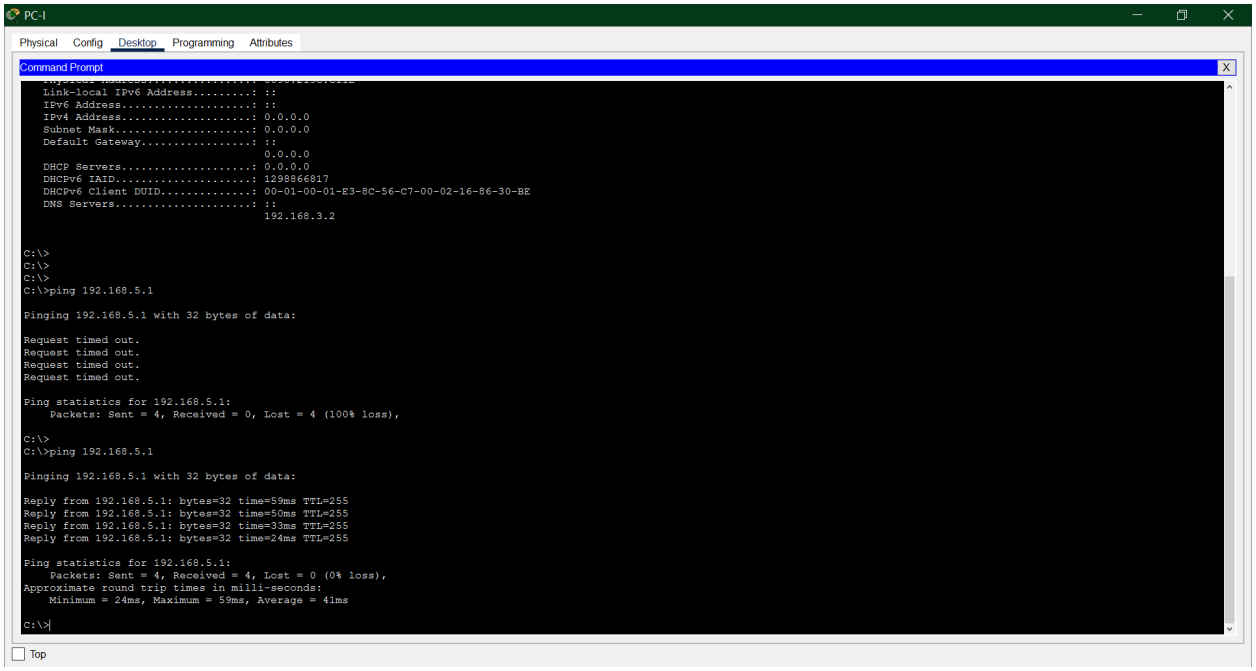


Figure 17: PC-I connectivity test before and after wireless reconfiguration.



9. NAT and PAT

9.1 Implementation and Purpose

Port Address Translation was set up on the London router for the Admin LAN. This ensures that the privately addressed devices on the `192.168.4.0/24` network are able to communicate through the router with its publicly accessible address.

PAT enables multiple private hosts to share a publicly available IP address by maintaining separate translations for individual connections. This reduces the number of public addresses that are needed for internal client devices.

The static NAT for the London servers was not completed and is therefore discussed separately as a limitation.

9.2 Configuration

NAT was used by the London Admin LAN when communicating through the London router. The appropriate internal and external interfaces were identified for NAT, and this allowed traffic from the private Admin LAN to be translated using the router's external address.

9.3 Testing and Evidence

PAT functionality was tested by generating traffic from the London Admin LAN and displaying the NAT translation table.

The output showed the internal host:

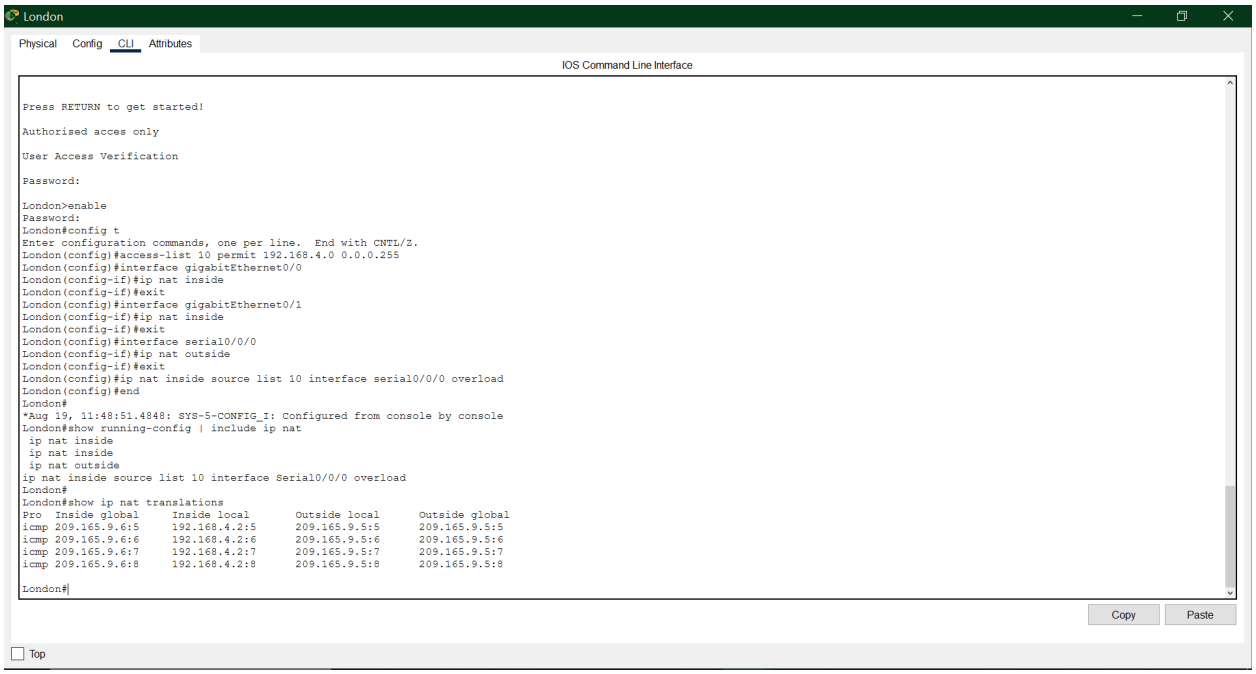
`192.168.4.2`

being translated to:

`209.165.9.6`

The presence of multiple ICMP translation entries confirmed that PAT was actively translating the generated traffic.

Figure 18: London PAT translation table.



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10. Overall Security Testing

Testing was carried out throughout the implementation rather than relying only on configuration commands. Both successful and deliberately unsuccessful communication attempts were used where appropriate.

SECURITY CONTROL	TEST PERFORMED	RESULT
Secure remote access / AAA	SSH authentication was tested from the London Admin LAN	Successful

OSPF authentication	The OSPF neighbour relationship was checked after message-digest authentication was configured	Neighbour returned to FULL state
Centralised logging	Router logging configuration and the central Syslog server were checked	Logs successfully received by the central server
NTP	NTP configuration and server connectivity were verified	Central NTP server successfully configured
Dublin inter-VLAN filtering	Traffic was sent from the Sales LAN to the Purchasing LAN	Traffic successfully blocked
Dublin ACL verification	ACL match counters were checked after generating blocked traffic	Deny rule recorded matching packets
WLAN security	PC-I connectivity was tested after the Beijing wireless security settings were changed	Initially failed; successful after wireless profile was updated
MAC filtering	PC-I was configured as an authorised wireless client using its Wireless0 MAC address	Authorised client successfully connected

PAT	Traffic was generated from the London Admin LAN and the NAT translation table was examined	PAT translation successfully created

These tests demonstrated that the controls put in place were operating properly and were not just included in the device configurations.

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11. Limitations and Additional Security Considerations

11.1 Dublin–Beijing Site-to-Site VPN

A site-to-site VPN between Dublin and Beijing was not successfully implemented. This control was supposed to secure traffic travelling between Beijing and the internal network traveling via the public internet.

When configuring this control, the Dublin router was found to have its Security technology package disabled. The required ISAKMP/IPsec functionality was therefore unavailable in the router's current Packet Tracer configuration.

In a complete implementation, a site-to-site IPsec VPN would be configured for securing traffic between these two locations.

11.2 Static NAT

Static NAT for the London servers could not be done. Static NAT will provide a fixed relationship between the server's internal private addresses and external addresses.

A successful deployment for PAT was done on the London Admin LAN, but a complete implementation would include configuring and testing of the static server translations.

11.3 Complete Dublin Firewall Policy

The Dublin gateway-security configuration was only partially implemented. The ACLs were successful in preventing communication between the Sales and Purchasing LANs but the complete service-specific filtering requirements for London and Beijing and the full anti-spoofing policy were not implemented.

A production implementation of the ACL policy would involve making sure the necessary services have permission and unauthorised and spoofed traffic gets denied.

11.4 Network-Wide Deployment

Certain security controls were tested on selected network infrastructure rather than being deployed consistently across every applicable device. A full production installation would apply appropriate hardening, monitoring, access-control and LAN-security policies consistently appropriately across the enitre infrastructure.

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12. Conclusion

The following controls have been added to the current three sites network in order to enhance its security. Administrative access was hardened using stronger device configuration, SSH and AAA. Monitoring was enhanced by centralised logging and time synchronisation. OSPF authentication secured routing communication. ACLs provided segmentation between the Dublin LANs. Wireless security was strengthened in Beijing and PAT was implemented for the London Admin network.

The testing demonstrated that several of these controls operated successfully, including authenticated routing, remote SSH access, PAT translations, wireless connectivity and blocked communication between inter-LAN traffic. There were some aspects in which the implementation was not successful, particularly the site-to-site VPN, static NAT and the full Dublin firewall policy. These areas would be priorities for further development in a production deployment.

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13. References

Technological University Dublin, Network Security Module Lab Materials, 2025/2026.

- Lab 2 Part 2 – Configure Packet Filter Firewalls to Mitigate Attacks
- Lab 3 Part 1 – Configuring AAA Authentication on Routers
- Lab 4 Part 2 – Configuring OSPF Authentication, NTP, Syslog and SSH
- Lab 5 Part 1 – Mitigating Against STP Manipulation & CAM Table Attacks.
- Lab 5 Part 2 – Mitigating Against VLAN Attacks
- Lab 6 Part 1 – Configuring Wireless Security
- Lab 6 Part 2 – Configure a Wireless Network for SOHO
- Lab 7 Part 1 – Configure Static NAT and Dynamic NAT
- Lab 7 Part 2 – Configuring Port Address Translation (PAT)
- Lab 8 Part 2 – Configure a Site-to-Site VPN